NANO-MAMBA U-NET: A SPATIO-TEMPORAL FRAMEWORK FOR CARDIAC MOTION ANALYSIS

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UNIVERSITI MALAYA
KUALA LUMPUR

2026

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RESEARCH PROJECT SUBMITTED TO THE FACULTY OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY UNIVERSITI MALAYA, IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE OF MASTER OF ARTIFICIAL INTELLIGENCE

2026

UNIVERSITI MALAYA

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NANO-MAMBA U-NET: A SPATIO-TEMPORAL FRAMEWORK FOR CARDIAC MOTION ANALYSIS

ABSTRACT

Cardiac Magnetic Resonance (CMR) imaging is widely used for quantitative assessment of cardiac anatomy and function. In automated cardiac analysis, accurate segmentation of the right ventricle (RV), myocardium (MYO), and left ventricle (LV) is an important prerequisite for deriving functional measurements such as end-diastolic volume, end-systolic volume, myocardial mass, and ejection fraction. Although U-Net-style convolutional neural networks remain strong foundations for medical image segmentation, practical deployment also requires attention to model size, inference cost, and reproducible validation. This thesis investigates Nano-Mamba U-Net, a lightweight Mamba-inspired 3D U-Net architecture for cardiac MRI segmentation. The model keeps the encoder-decoder structure and skip connections of U-Net, while inserting a compact sequence-processing bottleneck at the compressed latent representation. This bottleneck serializes a 3D spatial feature grid into a one-dimensional sequence, applies a lightweight gated module inspired by Mamba-style sequence modeling, and reshapes the output back into volumetric form. It does not model cardiac time: ED and ES volumes are processed independently, and tensor depth denotes anatomical slices. The implementation is therefore described as Mamba-inspired rather than as a full reproduction of the original hardware-aware selective scan algorithm.

The proposed model was evaluated on the Automated Cardiac Diagnosis Challenge (ACDC) dataset using a deterministic patient-level validation protocol. The final experiment used 80 training patients and 20 held-out validation patients, corresponding to 160 training cases and 40 validation cases. All models used the same preprocessing and validation procedure, although memory-constrained batch sizes differed. Nano-Mamba U-Net achieved a validation mean Dice Similarity Coefficient (DSC) of 84.78% with 1.46 million reported parameters. It improved over the 3D U-Net baseline, which achieved 80.83% mean DSC with 4.81 million parameters. However, SegResNet16 achieved the highest mean DSC of 86.70%, and the No-Mamba ablation achieved 85.64%. These results show that Nano-Mamba U-Net should not be claimed as the most accurate model by Dice. Instead, its main contribution is an accuracy-efficiency trade-off: it provides competitive held-out validation performance while using the smallest reported parameter count among the evaluated models. Post-hoc patient-level bootstrap intervals describe uncertainty on this validation cohort; independent test evaluation, multi-seed training, external validation, and true 4D cardiac motion modeling remain future work.

Keywords: Cardiac MRI Segmentation, Deep Learning, Mamba-Inspired Architecture, State Space Models, Accuracy-Efficiency Trade-off.

NANO-MAMBA U-NET: RANGKA KERJA RUANG-MASA UNTUK ANALISIS PERGERAKAN JANTUNG

ABSTRAK

Pengimejan Resonans Magnetik Jantung (CMR) digunakan secara meluas untuk penilaian kuantitatif anatomi dan fungsi jantung. Dalam analisis jantung automatik, segmentasi yang tepat bagi ventrikel kanan (RV), miokardium (MYO), dan ventrikel kiri (LV) merupakan langkah penting sebelum ukuran fungsi seperti isipadu akhir diastol, isipadu akhir sistol, jisim miokardium, dan pecahan ejeksi boleh diperoleh. Walaupun seni bina berasaskan U-Net masih menjadi asas yang kukuh dalam segmentasi imej perubatan, pelaksanaan praktikal juga perlu mempertimbangkan saiz model, kos inferens, dan protokol pengesahan yang boleh diulang. Tesis ini mengkaji Nano-Mamba U-Net, iaitu seni bina 3D U-Net ringan yang diinspirasikan oleh Mamba untuk segmentasi MRI jantung. Model ini mengekalkan struktur pengekod-penyahkod dan sambungan langkau U-Net, sambil memasukkan lapisan bottleneck pemprosesan jujukan yang padat pada perwakilan laten termampat. Lapisan ini menyusun grid ciri spatial 3D kepada jujukan satu dimensi, menggunakan modul berpagar ringan yang diinspirasikan oleh pemodelan jujukan gaya Mamba, dan membentuk semula output kepada bentuk volumetrik. Ia tidak memodelkan masa jantung: isipadu ED dan ES diproses secara berasingan, manakala dimensi kedalaman mewakili hirisan anatomi. Oleh itu, pelaksanaan ini diterangkan sebagai Mamba-inspired dan bukannya pelaksanaan penuh algoritma selective scan asal.

Model yang dicadangkan dinilai menggunakan set data Automated Cardiac Diagnosis Challenge (ACDC) dengan protokol pengesahan berasaskan pesakit. Eksperimen akhir menggunakan 80 pesakit untuk latihan dan 20 pesakit untuk pengesahan, bersamaan dengan 160 kes latihan dan 40 kes pengesahan. Semua model menggunakan prapemprosesan dan prosedur pengesahan yang sama, walaupun saiz kelompok berbeza kerana kekangan memori. Nano-Mamba U-Net mencapai min Dice Similarity Coefficient (DSC) pengesahan sebanyak 84.78% dengan 1.46 juta parameter yang dilaporkan. Model ini mengatasi garis dasar 3D U-Net, yang mencapai 80.83% min DSC dengan 4.81 juta parameter. Walau bagaimanapun, SegResNet16 mencapai min DSC tertinggi sebanyak 86.70%, manakala ablation No-Mamba mencapai 85.64%. Keputusan ini menunjukkan bahawa Nano-Mamba U-Net tidak harus didakwa sebagai model paling tepat berdasarkan Dice. Sebaliknya, sumbangan utamanya ialah keseimbangan antara ketepatan dan kecekapan: model ini memberikan prestasi pengesahan yang kompetitif dengan bilangan parameter dilaporkan paling kecil dalam kalangan model yang dinilai. Selang bootstrap pasca-hoc pada peringkat pesakit menerangkan ketidakpastian dalam kohort pengesahan ini; penilaian set ujian bebas, latihan berbilang benih, pengesahan luaran, dan pemodelan gerakan jantung 4D sebenar kekal sebagai kerja masa hadapan.

Kata Kunci: Segmentasi MRI Jantung, Pembelajaran Mendalam, Seni Bina Diinspirasikan Mamba, Model Ruang Keadaan, Keseimbangan Ketepatan-Kecekapan.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor, Dr. Uzair Iqbal, for his guidance and feedback throughout this project. I am also grateful to the Faculty of Computer Science and Information Technology, Universiti Malaya, for providing the academic environment and resources needed to complete this research. Finally, I would like to thank my family and friends for their continuous support and encouragement.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

1.1.1 Clinical Significance of Cardiovascular Diseases

Cardiovascular diseases (CVDs) remain one of the most important global causes of mortality. According to the World Health Organization (WHO), CVDs caused an estimated 19.8 million deaths in 2022, corresponding to approximately 32% of all global deaths (World Health Organization, 2025). This figure is not used in this thesis as a direct claim about the performance of any computational model. Rather, it establishes the clinical context in which efficient and reproducible cardiac image analysis tools are valuable. When the global disease burden is high, even small improvements in the reliability or scalability of cardiac assessment workflows can have practical importance.

Cardiac Magnetic Resonance (CMR) imaging is widely used for non-invasive assessment of cardiac anatomy and function (Bernard et al., 2018). From CMR images, clinicians can derive functional indices such as Ejection Fraction (EF), End-Diastolic Volume (EDV), End-Systolic Volume (ESV), and Myocardial Mass. These measurements depend on accurate delineation of the Right Ventricle (RV), Myocardium (MYO), and Left Ventricle (LV). Therefore, automated segmentation of these structures is a central step in computer-assisted cardiac analysis. In this project, segmentation is treated as a prerequisite for later cardiac functional analysis; the experiment itself does not calculate volumes, EF, or motion fields.

The clinical relevance of this task comes from the relationship between segmentation masks and quantitative cardiac indices. If V_{ED} denotes the ventricular volume at end-diastole and V_{ES} denotes the volume at end-systole, ejection fraction can be expressed as:

$$EF = \frac{V_{ED} - V_{ES}}{V_{ED}} \times 100\%. \tag{1.1}$$

This equation shows why segmentation errors matter. A small boundary error can change the estimated cavity volume, and the resulting volume error can propagate into EF. The equation motivates accurate masks, but it is not an evaluated output of the present experiment.

1.1.2 The Bottleneck of Manual Delineation and Traditional Methods

Manual delineation of cardiac structures requires expert knowledge and substantial effort. A cine CMR study may contain many slices and cardiac phases. The ACDC challenge paper emphasizes that accurate segmentation of RV, MYO, and LV is essential for cardiac diagnosis and that manual annotation can be affected by inter-observer variability (Bernard et al., 2018). Instead of relying on an unsupported fixed time estimate, this thesis treats manual annotation as a practical bottleneck because it is labor-intensive, repetitive, and difficult to scale across large datasets.

Manual segmentation is also not purely mechanical. The myocardium has an annular shape and may be difficult to separate from papillary muscles. The RV has a thin wall and variable geometry. The basal and apical slices often contain ambiguous anatomy because the visible structures gradually appear or disappear across the slice direction. These difficulties mean that even expert annotations may vary. Automated segmentation is therefore attractive not because it removes the need for clinical expertise, but because it can provide a consistent computational starting point for measurement and review.

Traditional image processing methods, including thresholding, active contours, level sets, and graph-cut algorithms, were historically applied to medical image segmentation. However, these methods rely heavily on hand-crafted features, manual initialization, or assumptions about image intensity distributions. In CMR images, boundary ambiguity, partial volume effects, papillary muscles, and scanner-dependent intensity variations make fully automated segmentation challenging (Chen et al., 2020). These limitations motivated the transition toward deep learning methods that learn task-specific representations from annotated data.

1.1.3 Deep Learning for Cardiac Segmentation

Deep learning shifted medical image segmentation from hand-crafted feature engineering toward data-driven representation learning. Fully Convolutional Networks (FCNs) introduced dense pixel-wise prediction for semantic segmentation (Long et al., 2015), while U-Net became a standard architecture for biomedical image segmentation due to its encoder-decoder design and skip connections (Ronneberger et al., 2015). For volumetric medical images, 3D U-Net extended this idea to three-dimensional convolutional processing (Çiçek et al., 2016). These architectures are important foundations for this thesis, and this project does not claim them as original contributions.

Despite their success, convolutional networks still rely on local kernels. A $3 \times 3 \times 3$ convolution extracts information from a local voxel neighbourhood. Stacking several convolutional layers increases the effective receptive field, but the operation remains locally constructed. In cardiac MRI, local detail and wider anatomical context are both important. The model must distinguish the myocardium from the blood pool at local boundaries while also maintaining enough contextual awareness to understand the relative positions of RV, MYO, and LV.

Transformer-based architectures address long-range interactions through self-attention (Vaswani et al., 2017; Dosovitskiy et al., 2021). Medical segmentation models such as UNETR and Swin UNETR demonstrate the usefulness of global or window-based token interactions in volumetric segmentation tasks (Hatamizadeh et al., 2022b,a). However, self-attention can increase memory and computational requirements, which motivates research into more efficient sequence modeling alternatives. This thesis does not argue that Transformers are unsuitable for medical imaging. Instead, it investigates whether a lighter sequence-inspired bottleneck can be useful when the hardware environment is constrained.

1.1.4 Motivation for a Mamba-Inspired Bottleneck

Selective State Space Models (SSMs), including Mamba, have recently attracted attention because they can model long sequences with linear-time complexity (Gu and Dao, 2023). This thesis investigates whether a lightweight Mamba-inspired sequence bottleneck can be integrated into a 3D U-Net-style architecture for cardiac MRI segmentation. The implemented model serializes compressed 3D volumetric features into a one-dimensional token sequence, applies a lightweight gated sequence module inspired by Mamba, and reshapes the result back into volumetric form.

The motivation is based on the accuracy-efficiency trade-off. A model with the highest Dice score is not necessarily the most suitable model for all environments if it requires more memory, more parameters, or longer inference time. Conversely, a very small model is not useful if it fails to segment clinically relevant structures. This thesis therefore evaluates Nano-Mamba U-Net not only by segmentation accuracy but also by parameter count and inference speed. The research question is whether a compact sequence-inspired bottleneck can produce competitive validation performance while maintaining a small model size.

The thesis title refers to cardiac motion analysis because segmentation of end-diastolic and end-systolic cardiac volumes is a prerequisite for computing functional indicators such as EDV, ESV, and EF. The registered title is retained, but the completed empirical scope is narrower than full motion analysis: ED and ES volumes are processed as separate

3D cases, and the tensor depth axis contains anatomical slices rather than time. The model does not jointly process a cine sequence, estimate pixel-wise displacement, enforce temporal consistency, or compute functional indices. Cardiac motion remains the clinical motivation and a future extension, while the evaluated contribution is 3D segmentation.

1.2 Scope of the Study

The scope of this research is defined as follows:

- **Dataset Boundary:** The empirical evaluation uses the Automated Cardiac Diagnosis Challenge (ACDC) dataset (Bernard et al., 2018). The segmentation targets are RV, MYO, and LV.
- **Task Boundary:** The project focuses on four-class 3D cardiac MRI segmentation of labelled ED and ES volumes processed independently. Slice depth is not time, and no explicit temporal or motion-vector estimation is evaluated.
- **Implementation Boundary:** The proposed Nano-Mamba U-Net uses a Mamba-inspired bottleneck implemented in PyTorch. It should not be interpreted as a full reproduction of the original hardware-aware selective scan implementation.
- **Evaluation Boundary:** The main reported results are based on a deterministic patient-level 80/20 validation split of the ACDC training cohort. The study does not claim performance on the hidden official ACDC test set.
- **Hardware Boundary:** Training, validation, and computational profiling were conducted on a single consumer-grade NVIDIA RTX 4060 GPU with 8GB VRAM. Reported FPS is specific to the implemented batch-one benchmark and is not treated as a universal property of an architecture.

These boundaries are deliberately stated to avoid over-claiming. The goal of the thesis is to produce a rigorous and reproducible project-level investigation, not to claim clinical deployment readiness or universal superiority over all existing cardiac segmentation methods.

1.3 Organization of the Thesis

This thesis is organized into eight chapters. Chapter 1 introduces the clinical and computational motivation. Chapter 2 defines the research problem. Chapter 3 states the objectives and research questions. Chapter 4 reviews relevant work in cardiac image segmentation, CNNs, attention models, residual networks, and state space models. Chapter 5 describes the dataset, preprocessing pipeline, architecture, training protocol, and evaluation method. Chapter 6 reports the validation results. Chapter 7 discusses the findings and limitations. Chapter 8 concludes the thesis and outlines future work.

CHAPTER 2: RESEARCH PROBLEM

Automated cardiac MRI segmentation is not only an accuracy problem. A clinically useful model must also be computationally practical and robust across different cardiac anatomies. This thesis focuses on the gap between accurate volumetric segmentation and lightweight implementation. The problem is framed around three connected issues: local context modeling, computational cost, and anatomical variability.

2.1 Local Context Modeling in Lightweight 3D CNNs

3D CNNs are effective for volumetric segmentation because they process spatial neighborhoods directly. However, a standard $3 \times 3 \times 3$ convolution only aggregates local voxel information at each layer. Although deeper networks can enlarge the effective receptive field, lightweight models with limited depth and channel capacity may still struggle to capture broader contextual relationships across the heart. This limitation is important in CMR segmentation because RV, MYO, and LV boundaries can be ambiguous at basal and apical slices.

The issue is not that convolution is ineffective. Convolution is one of the reasons U-Net-style models are successful in medical segmentation. The problem is that a lightweight model must compress information aggressively to remain small. When the deepest representation is too narrow or too local, the model may lose broader anatomical relationships. A useful bottleneck should therefore preserve compactness while allowing information exchange across the compressed spatial representation. This is the specific architectural motivation for inserting a sequence-processing block at the bottleneck rather than replacing the entire U-Net with a large attention model.

2.2 Computational Cost of Larger Segmentation Models

Stronger segmentation architectures often increase depth, width, attention operations, or residual blocks to improve representation capacity. This can improve accuracy but also increases parameters, memory usage, and inference cost. For resource-constrained clinical, educational, or prototype environments, a model with lower parameter count and acceptable accuracy may be preferable to a heavier model with marginally higher Dice scores. Therefore, the central problem is not simply to maximize Dice, but to study the trade-off between segmentation accuracy and computational efficiency.

This thesis uses parameter count and inference speed as supporting metrics because Dice alone does not describe the full engineering value of a model. A model with 1.46M parameters and competitive Dice may be useful even if it is not the highest-accuracy model. Conversely, if a lightweight model achieves lower Dice without meaningful savings, it would not be justified. The final evaluation therefore compares Nano-Mamba U-Net against baselines and ablation models using the same split and training protocol.

2.3 Pathological and Anatomical Variability

The ACDC dataset contains normal subjects and multiple cardiac pathology groups, including dilated cardiomyopathy, hypertrophic cardiomyopathy, myocardial infarction, and abnormal right ventricle cases (Bernard et al., 2018). These cases introduce variation in ventricular size, myocardial thickness, and boundary appearance. A segmentation model must therefore handle both normal cardiac geometry and pathological deformation.

Anatomical variability is particularly important for the RV and MYO. The RV has a complex shape and thinner wall, which makes its segmentation more sensitive to boundary ambiguity. The MYO is a narrow structure, so small absolute boundary errors can cause a noticeable Dice reduction. This explains why class-specific Dice is reported in the results chapter instead of reporting only one aggregate number. The evaluation must show whether the model performs consistently across RV, MYO, and LV.

2.4 Research Gap

Existing segmentation backbones such as 3D U-Net, Attention U-Net, and SegResNet provide strong baselines, but they represent different points on the accuracy-efficiency spectrum. The research gap addressed in this thesis is whether a compact Mamba-inspired bottleneck can provide competitive validation performance with fewer parameters than conventional or residual alternatives, while remaining feasible on a consumer GPU.

The gap is not a claim that previous methods are inadequate in all settings. SegResNet, for example, remains a strong residual convolutional baseline. The gap is narrower and more appropriate for this project: there is value in testing a compact sequence-inspired bottleneck under a rigorous patient-level validation protocol, especially when earlier exploratory scripts may overestimate performance by evaluating on non-held-out data. This thesis therefore emphasizes reproducibility and honest comparison as much as architectural novelty.

CHAPTER 3: RESEARCH OBJECTIVES

The aim of this research is to design, implement, and evaluate a lightweight Mamba-inspired U-Net architecture for cardiac MRI segmentation. The study emphasizes empirical validation, parameter efficiency, and transparent comparison with established segmentation baselines. The objectives are designed to connect directly to the experiments reported in Chapter 6.

3.1 Specific Objectives

1. To implement a Nano-Mamba U-Net architecture that inserts a lightweight Mamba-inspired sequence bottleneck into a 3D U-Net-style encoder-decoder network.
2. To evaluate the segmentation performance of the proposed model on RV, MYO, and LV using Dice Similarity Coefficient on a deterministic patient-level validation split of the ACDC training cohort.
3. To compare the proposed model with 3D U-Net, Attention U-Net, SegResNet, and ablation variants using the same split, preprocessing, epoch budget, optimizer family, and validation protocol, with model-specific batch sizes disclosed as a limitation.
4. To quantify computational efficiency using reported trainable-parameter count and batch-one inference speed.

The first objective is implemented through the model architecture described in Chapter 5. The second and third objectives are addressed through the quantitative validation table in Chapter 6. The fourth objective is addressed by reporting trainable parameters and inference speed together with Dice. This mapping is included to make the thesis structure explicit and to ensure that the stated objectives are actually answered by the experimental design.

3.2 Research Questions

1. How can compressed 3D cardiac MRI feature maps be serialized and processed by a Mamba-inspired bottleneck within a U-Net-style segmentation architecture?
2. Does the proposed Nano-Mamba U-Net achieve competitive held-out validation Dice compared with 3D U-Net, Attention U-Net, SegResNet, and ablation variants?
3. What accuracy-efficiency trade-off is observed when comparing Nano-Mamba U-Net with heavier or non-Mamba alternatives in terms of Dice score, parameter count, and inference speed?

These research questions are intentionally phrased in a measurable way. The first question is answered by architecture implementation rather than by a numerical score alone. The second question is answered by held-out validation Dice. The third question is answered by jointly interpreting Dice, parameter count, and speed. This structure avoids unsupported claims such as absolute superiority and keeps the thesis aligned with the actual experimental evidence.

CHAPTER 4: LITERATURE REVIEW

This chapter reviews the scientific and technical literature that supports the design choices of this thesis. The purpose is not to claim that the proposed model replaces all existing cardiac segmentation methods. Instead, the review identifies the specific research context in which a lightweight Mamba-inspired U-Net is meaningful: automated cardiac MRI segmentation requires accurate anatomical delineation, but practical models must also consider parameter count, memory usage, and reproducible validation.

4.1 Cardiac MRI Segmentation as a Clinical Computing Problem

Cardiac Magnetic Resonance (CMR) imaging is widely used for non-invasive assessment of cardiac morphology and function. In the ACDC benchmark, the segmentation targets are the Right Ventricle (RV), Myocardium (MYO), and Left Ventricle (LV), which are clinically important because they support the computation of ventricular volumes, myocardial mass, and ejection fraction (Bernard et al., 2018). These quantities are not obtained directly from a neural network classifier; they require spatial masks from which anatomical volumes can be computed. Therefore, segmentation quality is central to the reliability of downstream cardiac function analysis.

CMR segmentation is challenging because the difficulty is not uniform across all structures. LV segmentation is often more stable because the LV cavity is relatively compact and has a clearer blood-pool boundary. In contrast, the myocardium is a thin ring-like structure whose boundaries can be affected by partial volume effects, papillary muscles, and signal ambiguity. The RV is also difficult because it has a more variable crescent-like shape and thinner wall. These anatomical observations explain why a single aggregate Dice score is insufficient; class-specific performance should also be reported when evaluating cardiac segmentation models.

The ACDC dataset is useful for this thesis because it includes multiple cardiac conditions, including normal subjects and pathological groups such as dilated cardiomyopathy, hypertrophic cardiomyopathy, myocardial infarction, and abnormal right ventricle cases (Bernard et al., 2018). This heterogeneity makes the benchmark more relevant than a dataset containing only healthy cardiac anatomy. Nevertheless, the ACDC training cohort remains a limited dataset, so patient-level separation between training and validation is essential. If frames from the same patient appear in both training and evaluation, the reported performance may reflect memorization of patient-specific anatomy rather than generalization.

4.2 Traditional Medical Image Segmentation Methods

Before deep learning became dominant, medical image segmentation commonly relied on intensity thresholding, deformable models, graph-based optimization, and level-set methods. These approaches remain important historically because they clarify why learning-based segmentation became attractive.

A simple thresholding method assumes that the target object can be separated from the background by intensity. Otsu's method, for example, selects a threshold T by maximizing the between-class variance (Otsu, 1979). If the two classes have probabilities $\omega_0(T)$ and $\omega_1(T)$, and means $\mu_0(T)$ and $\mu_1(T)$, the objective can be written as:

$$\sigma_B^2(T) = \omega_0(T)\omega_1(T)\left[\mu_0(T) - \mu_1(T)\right]^2. \tag{4.1}$$

This formulation is elegant because it does not require annotated training data. However, it is not sufficient for cardiac MRI segmentation. MRI intensities are not standardized in the same way as CT Hounsfield Units, and myocardium, blood pool, and surrounding tissues may overlap in intensity. As a result, a global threshold cannot reliably separate RV, MYO, and LV across patients.

Active contour models, introduced by Kass et al., formulate segmentation as an energy minimization problem (Kass et al., 1988). A contour evolves under internal smoothness forces and external image forces. A simplified energy functional can be written as:

$$E_{snake} = \int_0^1 \left[\frac{1}{2}\alpha|v'(s)|^2 + \frac{1}{2}\beta|v''(s)|^2 + E_{image}(v(s)) \right] ds. \tag{4.2}$$

Here, the first two terms regulate contour continuity and smoothness, while E_{image} attracts the contour to image features such as edges. This is useful when boundaries are clear, but cardiac MRI boundaries can be weak or ambiguous at basal and apical slices. Moreover, deformable models often require careful initialization, which limits their practicality for fully automated pipelines.

Level-set methods represent the contour implicitly as the zero level of a higher-dimensional function ϕ (Osher and Sethian, 1988). A typical evolution equation is:

$$\frac{\partial \phi}{\partial t} + F|\nabla\phi| = 0. \tag{4.3}$$

The implicit representation allows topology changes more naturally than parametric snakes. However, iterative numerical optimization and sensitivity to image forces remain practical limitations. For this reason, traditional methods are useful background, but they do not remove the need for robust data-driven segmentation in heterogeneous CMR datasets (Chen et al., 2020).

4.3 Deep Learning Segmentation Architectures

4.3.1 Fully Convolutional Networks and U-Net

Fully Convolutional Networks (FCNs) showed that classification-style neural networks could be reformulated for dense pixel-wise prediction by replacing fully connected layers with convolutional operations (Long et al., 2015). This idea was important because segmentation requires a prediction for every pixel or voxel rather than one image-level label.

U-Net became one of the most influential architectures in biomedical image segmentation (Ronneberger et al., 2015). Its main contribution is the encoder-decoder structure with skip connections. The encoder gradually reduces spatial resolution to learn semantic features, while the decoder restores resolution to produce a segmentation mask. Skip connections transfer high-resolution spatial features from the encoder to the decoder, which helps recover fine boundaries. If $\mathbf{X}_{enc}^{(l)}$ is an encoder feature map and $\mathbf{X}_{dec}^{(l+1)}$ is a decoder feature map from a coarser level, a simplified skip-connection operation can be written as:

$$\mathbf{X}_{dec}^{(l)} = \mathscr{F}\left(\mathbf{X}_{enc}^{(l)} \oplus \mathscr{U}\left(\mathbf{X}_{dec}^{(l+1)}\right)\right), \tag{4.4}$$

where $\mathscr{U}$ denotes up-sampling, $\oplus$ denotes channel-wise concatenation, and $\mathscr{F}$ denotes convolutional refinement. This equation explains why U-Net is suitable for medical segmentation: the decoder does not reconstruct boundaries only from compressed semantic features; it also receives high-resolution information from earlier encoder layers.

For volumetric data, 3D U-Net extends the same principle from 2D images to 3D tensors (Çiçek et al., 2016). V-Net also used volumetric convolution and Dice-based optimization for 3D medical segmentation (Milletari et al., 2016). These works are directly relevant to this thesis because the proposed Nano-Mamba U-Net uses a 3D U-Net-style encoder-decoder as its architectural base. The contribution of this thesis is not the invention of the U-Net framework itself. Rather, the project studies whether the bottleneck of such a framework can be modified into a compact sequence-processing module.

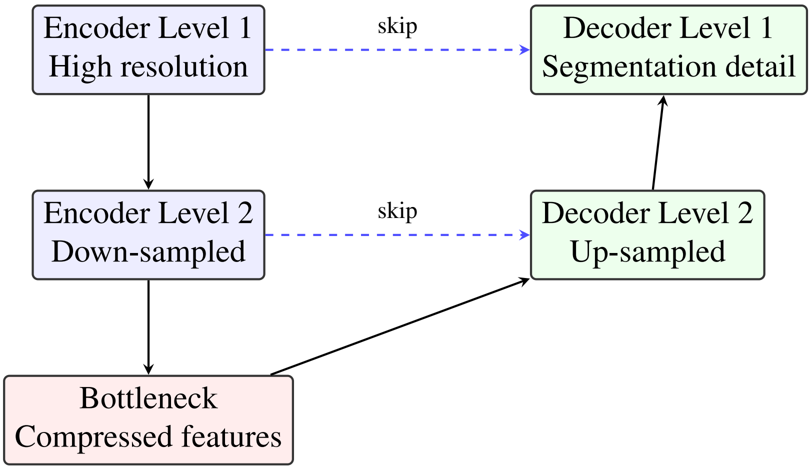

Figure 4.1: Conceptual encoder-decoder structure of U-Net-style segmentation models. Encoder blocks learn increasingly abstract features, decoder blocks recover spatial resolution, and skip connections preserve boundary information.

4.3.2 Attention U-Net

Attention U-Net introduced attention gates to suppress irrelevant encoder features before they are passed through skip connections (Oktay et al., 2018). This is relevant in medical images because large regions of an image may contain background structures unrelated to the target anatomy. In a simplified attention gate, an encoder feature x and a decoder gating signal g are projected into a shared representation and transformed into an attention coefficient:

$$\alpha = \sigma\left(\psi^T \delta\left(W_x x + W_g g + b\right)\right). \tag{4.5}$$

The filtered feature can then be represented as:

$$\hat{x} = \alpha \odot x. \tag{4.6}$$

Here, σ is a sigmoid function, δ is a nonlinear activation, and $\odot$ denotes element-wise multiplication. The key idea is not that attention automatically guarantees better segmentation, but that it provides a mechanism for feature selection. In the final experiment of this thesis, Attention U-Net did not outperform the other models, which illustrates why empirical comparison under the same split is necessary.

4.3.3 Residual Networks and SegResNet

Residual learning was introduced to make deeper networks easier to optimize (He et al., 2016). Instead of asking stacked layers to learn a full mapping $\mathscr{H}(x)$, a residual block learns a residual function $\mathscr{F}(x)$ and adds the input back to the output:

$$y = \mathscr{F}(x) + x. \tag{4.7}$$

This identity shortcut improves gradient flow and can make deeper convolutional models more trainable. SegResNet-style medical segmentation architectures use residual blocks in an encoder-decoder setting (Myronenko, 2019). In this thesis, the MONAI SegResNet configuration with 16 initial filters serves as a residual convolutional alternative to the proposed Nano-Mamba U-Net. The final results show that SegResNet16 achieved the highest mean Dice, which must be acknowledged rather than hidden.

4.4 Transformers and Sequence Modeling in Medical Imaging

Transformers introduced self-attention as a way to model long-range dependencies (Vaswani et al., 2017). Vision Transformer adapted this idea to image classification by treating image patches as tokens (Dosovitskiy et al., 2021). In medical imaging, UNETR used transformer encoders for 3D segmentation (Hatamizadeh et al., 2022b), while Swin Transformer and Swin UNETR introduced hierarchical shifted-window attention to reduce the cost of global attention (Liu et al., 2021; Hatamizadeh et al., 2022a).

The computational concern with standard self-attention is that the attention matrix grows quadratically with the number of tokens. For an input sequence of length N, the attention matrix has size $N \times N$, and the simplified attention operation is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V. \tag{4.8}$$

This equation shows why sequence length matters: the multiplication QK^T compares every token with every other token. Window-based methods reduce this cost, but the broader issue remains important for volumetric segmentation because 3D images can produce many spatial tokens.

It would be inaccurate to claim that all transformer models are infeasible or that they are unsuitable for medical imaging. Prior work shows that transformer-based segmentation can be effective. The more careful motivation for this thesis is narrower: for a master's project focused on lightweight deployment and reproducible experiments on a consumer GPU, it is reasonable to investigate alternatives that attempt to combine broader context modeling with lower parameter count.

4.5 State Space Models and Mamba-Inspired Design

State Space Models (SSMs) represent sequence dynamics through a hidden state. A continuous-time linear state space model can be written as:

$$h'(t) = Ah(t) + Bx(t), \qquad y(t) = Ch(t). \tag{4.9}$$

Here, $h(t)$ is the hidden state, $x(t)$ is the input signal, $y(t)$ is the output signal, and A, B, and C are state transition and projection matrices. In the context of deep learning, the appeal of SSMs is that they can model long sequences without explicitly constructing a full $N \times N$ attention matrix.

Mamba extends this line of work through selective state space modeling and hardware-aware sequence processing (Gu and Dao, 2023). Recent biomedical segmentation studies have also explored Mamba-style modules for medical image segmentation, including U-Mamba for long-range dependency modeling and SegMamba for 3D medical segmentation (Ma et al., 2024; Xing et al., 2024). These works support the broader relevance of SSM-inspired modeling in medical image analysis, but they should be treated as related work rather than as evidence that every Mamba-inspired block will automatically improve cardiac segmentation.

However, this thesis must make a clear distinction between the original Mamba implementation, recent Mamba-based medical segmentation studies, and the implemented project model. The proposed Nano-Mamba U-Net uses a Mamba-inspired bottleneck implemented in PyTorch. It serializes compressed 3D feature maps into a sequence and applies a lightweight gated sequence-processing block. It is therefore more accurate to describe the contribution as a Mamba-inspired architectural exploration, not as a full reproduction of the original selective scan mechanism.

This distinction matters for scientific integrity. If the implementation does not include the full selective scan algorithm, the thesis should not claim that it has implemented the complete Mamba architecture. The contribution is still valid, but it should be framed correctly: the project investigates whether a compact sequence-inspired bottleneck can provide a useful accuracy-efficiency trade-off in cardiac MRI segmentation.

4.6 Loss Functions and Evaluation Metrics

Medical segmentation is often class-imbalanced because background voxels dominate the image volume, while structures such as myocardium occupy a smaller region. Dice-based objectives are commonly used because they directly optimize overlap between prediction and ground truth (Milletari et al., 2016; Sudre et al., 2017). For a predicted binary mask P and ground-truth mask G, the Dice Similarity Coefficient is:

$$\mathrm{DSC}(P,G) = \frac{2|P \cap G|}{|P| + |G|}. \tag{4.10}$$

In multi-class segmentation, Dice can be computed separately for RV, MYO, and LV, and then averaged. This is the reporting strategy used in this thesis.

Cross-entropy is also widely used because it provides voxel-wise classification supervision. If y_i is the true class at voxel i and $p_{i,c}$ is the predicted probability for class c, the multi-class cross-entropy can be expressed as:

$$\mathscr{L}_{CE} = -\sum_i \sum_c \mathbf{1}(y_i = c) \log p_{i,c}. \tag{4.11}$$

The final experiment uses MONAI's Dice-Cross Entropy loss, combining region-overlap supervision with voxel-wise classification supervision. This choice is practical rather than novel; it follows established segmentation practice and is supported by the MONAI framework (Cardoso et al., 2022).

Evaluation should also avoid leakage. If the same patient contributes samples to both training and validation, the measured Dice may overestimate generalization. Therefore, the final experimental design of this thesis uses patient-level splitting. This methodological point is as important as the architecture itself, because a strong model comparison is only meaningful when all models are trained and evaluated under the same split.

4.7 Summary of the Literature Gap

The literature shows that U-Net-style models remain strong foundations for biomedical segmentation, attention mechanisms can improve feature selection, residual models provide strong convolutional baselines, and transformer/SSM approaches offer ways to model broader contextual relationships. Recent Mamba-style medical segmentation studies further suggest that sequence and state-space ideas are becoming relevant in this area. The gap addressed by this thesis is not the absence of segmentation models, but the need to study a compact architecture under a rigorous and reproducible validation protocol.

Based on this review, the proposed Nano-Mamba U-Net is positioned as a lightweight Mamba-inspired extension of a 3D U-Net-style segmentation model. The appropriate research question is therefore not whether it universally surpasses all previous methods, but whether it offers a useful accuracy-efficiency trade-off for cardiac MRI segmentation on the ACDC dataset.

CHAPTER 5: RESEARCH DESIGN AND METHODOLOGY

This chapter describes the research design used to implement and evaluate the proposed Nano-Mamba U-Net. The methodology is written to make the experiment reproducible: it specifies the dataset boundary, data discovery procedure, patient-level split, preprocessing pipeline, model architecture, loss function, checkpoint selection, evaluation metrics, and computational profiling. The chapter also distinguishes clearly between methods adopted from prior work and components implemented in this project.

5.1 Overall Research Workflow

The study follows an experimental computing workflow. First, the ACDC cardiac MRI data are discovered and paired with corresponding segmentation labels. Second, patients are separated into training and validation groups using a deterministic patient-level split. Third, all models use the same preprocessing, patient split, epoch budget, optimizer family, and validation procedure, while memory-constrained batch sizes differ by model as reported below. Finally, the best checkpoint for each model is selected by validation mean Dice and evaluated descriptively on the same held-out validation patients.

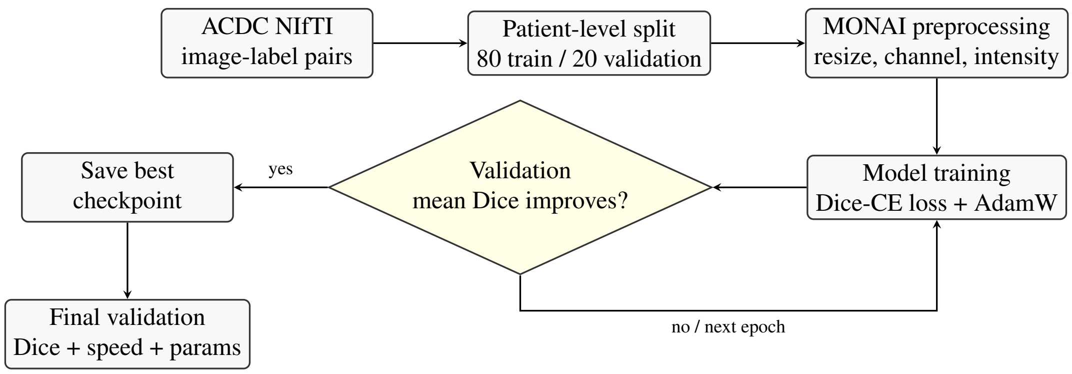

Figure 5.1: Reproducible experimental workflow. The main methodological safeguard is patient-level separation before training, so that validation cases come from patients not seen during optimization.

5.2 Dataset and Data Discovery

The experiment uses the Automated Cardiac Diagnosis Challenge (ACDC) dataset (Bernard et al., 2018). The task is multi-class segmentation of three anatomical structures: RV, MYO, and LV. The final discovery snapshot records 200 non-empty image-label path pairs from 100 patients. Each patient contributes labelled end-diastolic and end-systolic frames, giving two labelled 3D cases per patient.

A practical issue in this project is that the local dataset layout contains paths ending in `.nii` that may act as containers rather than directly readable NIfTI files. Therefore, the data discovery procedure checks whether each candidate path is a non-empty file or a directory containing a non-empty NIfTI path. Empty placeholders are skipped. It also requires exactly two labelled cases for every patient from `patient001` through `patient100`; duplicate case identifiers or incomplete patient coverage cause the pipeline to stop before splitting. This prevents a partially discovered dataset from silently entering the experiment.

5.3 Patient-Level Training and Validation Split

The split was performed at patient level rather than case level. This means all labelled frames from the same patient belong either to training or to validation, never both. This design prevents leakage of patient-specific anatomy into the validation set. With random seed 42 and an 80/20 split, the experiment used 80 training patients and 20 validation patients, corresponding to 160 training cases and 40 validation cases.

Let $\mathscr{P}$ denote the set of all patient identifiers. The split can be expressed as:

$$\mathscr{P}_{train} \cap \mathscr{P}_{val} = \varnothing, \qquad \mathscr{P}_{train} \cup \mathscr{P}_{val} = \mathscr{P}. \tag{5.1}$$

This condition is the core methodological safeguard of the experiment. It ensures that the validation Dice measures performance on unseen patients rather than on additional frames from patients used during training.

5.4 Preprocessing Pipeline

All models use the same deterministic preprocessing pipeline implemented with MONAI (Cardoso et al., 2022). The exact executed order is:

1. `LoadImaged` loads the paired NIfTI intensity volume and integer label volume.
2. `EnsureChannelFirstd` changes an array shaped $H \times W \times D$ into $1 \times H \times W \times D$.
3. `Resized` maps both arrays to $256 \times 256 \times 16$: trilinear interpolation is used for the continuous MRI intensity and nearest-neighbour interpolation for categorical labels.
4. `ScaleIntensityd` linearly maps the minimum and maximum intensity of the single-channel volume to 0 and 1.
5. `ToTensord` converts both arrays to PyTorch tensors for the data loader.

No random crop, rotation, elastic deformation, intensity augmentation, or other data augmentation is applied. Training and validation use the same deterministic geometric and intensity transforms; only the training data loader shuffles case order. Applying the same preprocessing to all models is necessary because otherwise observed performance differences could be caused by data handling rather than architecture.

The executed transform resizes array dimensions directly; it does not explicitly canonicalize image orientation or resample every case to a common physical voxel spacing before resizing. It also does not crop around the heart or restore predictions to native image geometry. Therefore, training and Dice evaluation occur on the common resized array grid. The reported experiment is a fixed-array-shape comparison rather than a native-space or spacing-preserving geometric evaluation. This preprocessing is documented rather than changed retrospectively, because changing it would require retraining every model and would produce a different experiment.

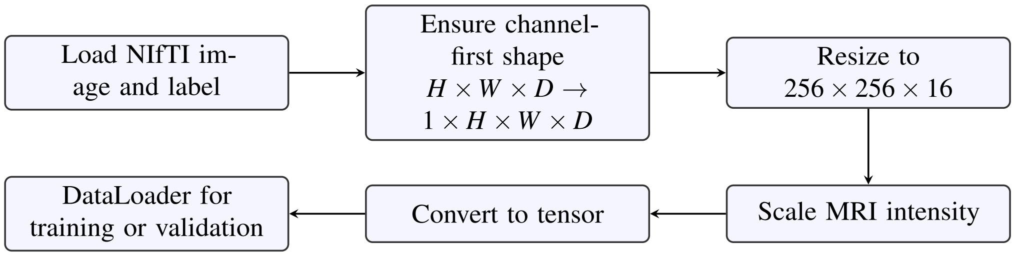

Figure 5.2: Preprocessing pipeline used for all compared models. Images are interpolated continuously, while labels are resized with nearest-neighbour interpolation to preserve class identities.

For continuous MRI intensity volumes, trilinear interpolation is used during resizing. If $V(x,y,z)$ denotes the interpolated intensity at a continuous coordinate, the interpolation can be described as a weighted combination of its eight neighbouring voxels:

$$V(x,y,z) \approx \sum_{i=0}^{1} \sum_{j=0}^{1} \sum_{k=0}^{1} w_{ijk} V_{ijk}. \tag{5.2}$$

Here, V_{ijk} are neighbouring voxel intensities and w_{ijk} are interpolation weights determined by the relative distance between the target coordinate and the neighbouring grid points. For segmentation labels, nearest-neighbour interpolation is used instead, because labels are categorical class identifiers rather than continuous intensity values.

MRI intensity values are relative and scanner-dependent rather than standardized physical units (Nyúl and Udupa, 1999). Therefore, intensity scaling is applied before network input. For a non-constant single-channel volume, the executed min-max mapping can be written as:

$$S(I_{xyz}) = \frac{I_{xyz} - \min(I)}{\max(I) - \min(I)}. \tag{5.3}$$

This maps the volume minimum to 0 and maximum to 1; MONAI handles the constant-volume edge case. The scaling is applied to each loaded volume, not across the full dataset. No histogram standardization or scanner-specific normalization is performed.

5.5 Nano-Mamba U-Net Architecture

The proposed model follows a 3D U-Net-style encoder-decoder structure. The encoder uses convolutional blocks and pooling to reduce spatial resolution while increasing feature channels. The decoder uses transposed convolutions and skip connections to recover spatial resolution and preserve boundary information. The architectural modification studied in this thesis is the insertion of a lightweight Mamba-inspired bottleneck at the compressed latent representation. For Nano-Mamba U-Net, the main tensor path is $B \times 1 \times 256 \times 256 \times 16$ at input, $B \times 128 \times 32 \times 32 \times 2$ after the third pooling operation and the first bottleneck convolution, and $B \times 4 \times 256 \times 256 \times 16$ at the output. The bottleneck therefore serializes $32 \times 32 \times 2 = 2048$ spatial tokens, each with 128 channels.

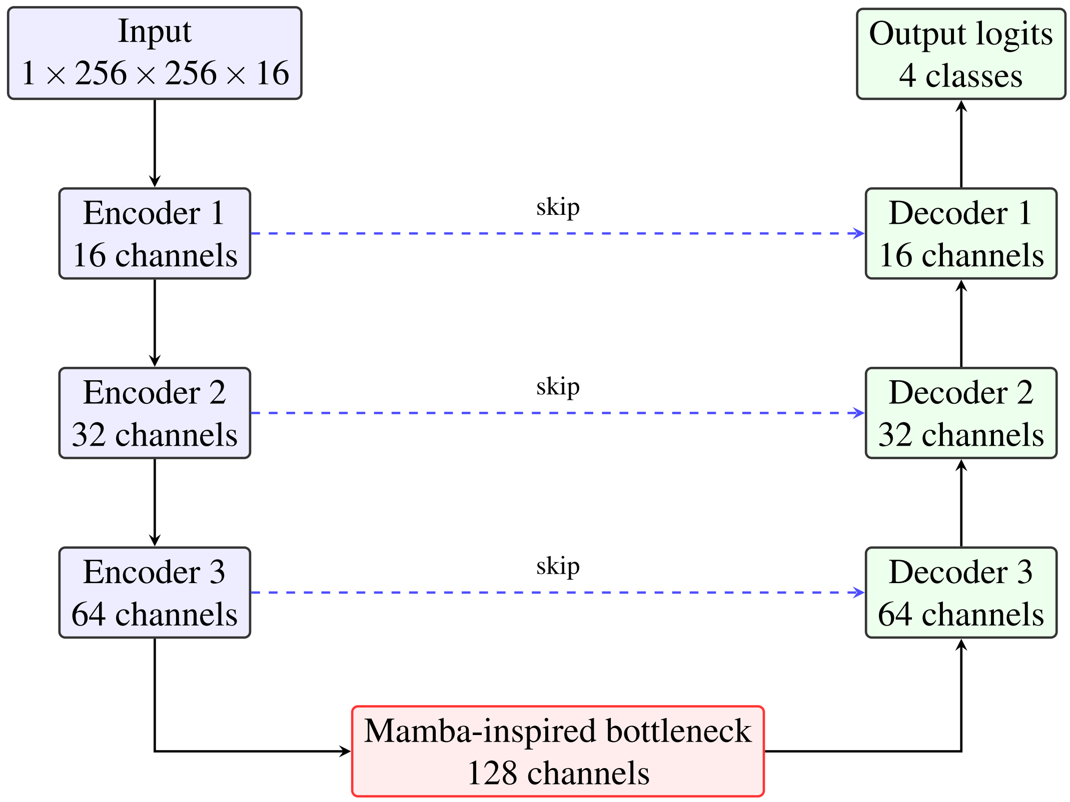

Figure 5.3: Nano-Mamba U-Net architecture. The model keeps the U-Net encoder-decoder and skip-connection structure, while augmenting the deepest convolutional bottleneck with a compact Mamba-inspired gated sequence module.

Each convolutional block consists of two 3D convolutional layers followed by batch normalization and ReLU activation. A convolutional layer transforms a local neighbourhood into an output feature using learned kernels. For a 3D convolution, this can be written as:

$$Y_{c_o}(p) = \sum_{c_i} \sum_{q \in \mathscr{K}} W_{c_o,c_i}(q) X_{c_i}(p+q) + b_{c_o}. \tag{5.4}$$

Here, X is the input feature map, Y is the output feature map, $\mathscr{K}$ is the local 3D kernel support, and W and b are trainable parameters. This equation also shows the locality of convolution: each output voxel depends on a neighbourhood around position p.

Batch normalization stabilizes training by normalizing intermediate features within a mini-batch (Ioffe and Szegedy, 2015). For a feature value x, batch normalization can be expressed as:

$$\hat{x} = \frac{x - \mu_{\mathscr{B}}}{\sqrt{\sigma^2_{\mathscr{B}} + \varepsilon}}, \qquad y = \gamma \hat{x} + \beta. \tag{5.5}$$

The parameters γ and β are learned, allowing the network to recover suitable feature scales after normalization. ReLU activation is then applied as:

$$f(x) = \max(0, x). \tag{5.6}$$

These components are standard neural network operations and are not claimed as novel contributions of this thesis.

5.6 Mamba-Inspired Bottleneck Implementation

The bottleneck receives a compressed 3D feature map. Let this tensor be:

$$\mathscr{X} \in \mathbb{R}^{B \times C \times H \times W \times D}. \tag{5.7}$$

The tensor is serialized across the spatial dimensions to obtain a sequence:

$$\mathbf{X}_{seq} = \text{reshape}(\mathscr{X}) \in \mathbb{R}^{B\times L\times C}, \qquad L = HWD. \tag{5.8}$$

This operation does not create new anatomical or temporal information. It uses a fixed raster order over the spatial grid so that the bottleneck can process compressed volumetric features as a sequence. Here, D is anatomical slice depth, not cardiac time; ED and ES volumes are processed as separate cases.

The implemented bottleneck is Mamba-inspired rather than a full selective scan implementation. It uses linear projection, a symmetric depthwise one-dimensional convolution, a scalar sigmoid gate per token, output projection, and residual addition. The executed graph can be represented as:

$$[\mathbf{U},\mathbf{V}] = \text{split}\big(\text{Proj}_{in}(\mathbf{X}_{seq})\big), \tag{5.9}$$

$$\mathbf{H} = \text{SiLU}(\text{DWConv1D}(\mathbf{U})), \qquad \mathbf{G} = \sigma\big(\text{Proj}_{gate}(\mathbf{H})_{:,:,1}\big), \tag{5.10}$$

$$\mathbf{Y} = \text{Proj}_{out}(\mathbf{H}\odot\mathbf{G}\odot\text{SiLU}(\mathbf{V})), \qquad \mathbf{Z} = \mathbf{Y} + \mathbf{X}_{seq}. \tag{5.11}$$

Here, Proj_{in} and Proj_{out} are learned linear projections, DWConv1D is the depthwise sequence convolution, σ is the sigmoid function, and $\odot$ denotes element-wise multiplication. In the executed implementation, Proj_{gate} produces $2d_{state}+1$ channels but only its first channel enters $\mathbf{G}$. The remaining legacy projection channels are retained for checkpoint compatibility but have no path to the output. The output sequence is reshaped back into 3D feature form before being passed to the decoder.

This design choice must be interpreted carefully. The original Mamba paper proposes selective state space sequence modeling (Gu and Dao, 2023). The implementation in this project uses a lightweight PyTorch block inspired by this direction, but it does not claim to reproduce the full hardware-aware selective scan. The research contribution is therefore the integration and evaluation of a compact sequence-inspired bottleneck inside a cardiac MRI segmentation model.

5.7 Comparison Models and Ablation Design

The experiment compares Nano-Mamba U-Net with several reference architectures. The 3D U-Net baseline represents the standard encoder-decoder approach. Attention U-Net represents attention-gated skip connections. SegResNet16 represents a residual convolutional segmentation model based on the SegResNet family (Myronenko, 2019). Two ablation variants are included to study the role of the bottleneck: No-Mamba U-Net replaces the Mamba-inspired bottleneck with convolutional blocks, while Half-Mamba U-Net reduces the bottleneck capacity.

The ablations are not parameter matched: the No-Mamba variant contains 2.29M reported parameters, whereas Nano-Mamba U-Net contains 1.46M. It also uses a different two-block convolutional bottleneck. The comparison can therefore describe the observed architecture-level accuracy-efficiency trade-off, but it cannot isolate a causal benefit or harm from sequence gating alone. This distinction is important because No-Mamba U-Net slightly outperformed Nano-Mamba U-Net in mean Dice, while Nano-Mamba U-Net retained the lowest reported parameter count.

5.8 Training Objective and Optimization

The models are trained using a Dice-Cross Entropy objective. Dice loss addresses region-level overlap, while cross entropy provides voxel-wise class supervision. For class c, the Dice score is:

$$\mathrm{DSC}_c = \frac{2\sum_i p_{i,c} g_{i,c}}{\sum_i p_{i,c} + \sum_i g_{i,c} + \varepsilon}. \tag{5.12}$$

The corresponding Dice loss can be written as:

$$\mathscr{L}_{Dice} = 1 - \frac{1}{C}\sum_{c=1}^{C} \mathrm{DSC}_c. \tag{5.13}$$

The multi-class cross-entropy term is:

$$\mathscr{L}_{CE} = -\sum_{i}\sum_{c} g_{i,c}\log(p_{i,c}). \tag{5.14}$$

The total loss used for optimization is:

$$\mathscr{L}_{total} = \mathscr{L}_{Dice} + \mathscr{L}_{CE}. \tag{5.15}$$

Here, $p_{i,c}$ denotes the predicted probability for voxel i and class c, while $g_{i,c}$ denotes the one-hot ground-truth label. This loss is a standard practical choice for medical segmentation and is implemented using MONAI's Dice-Cross Entropy loss. The executed constructor sets `to_onehot_y=True` and `softmax=True` but does not override `include_background`; under the documented MONAI default, the four-channel training Dice term includes background. By contrast, the reported evaluation Dice is computed only for the three foreground structures. The optimization loss and the reported metric therefore use related but not identical class averaging.

Optimization is performed with AdamW (Loshchilov and Hutter, 2019). The learning rate is 1×10^{-4} and weight decay is 1×10^{-5}. All models are trained for 150 epochs without a learning-rate scheduler or early stopping; validation is run after every epoch. UNet3D, Nano-Mamba, No-Mamba, and Half-Mamba use batch size 2; Attention U-Net and SegResNet16 use batch size 1 because of memory constraints. This is not a strictly controlled batch-size comparison. Different batch sizes imply different numbers of optimizer updates per epoch, and normalization choices also differ across model families. The table should therefore be interpreted as a comparison of executed model configurations, not a perfectly controlled isolation of architecture alone.

5.9 Checkpoint Selection and Evaluation Metrics

A key methodological correction in the final experiment is validation-based checkpoint selection. Earlier exploratory scripts saved best models according to training loss, which is not a reliable indicator of generalization. In the final experiment, the checkpoint is saved only when validation mean Dice improves. If $\overline{D}_{val}^{(e)}$ denotes the validation mean Dice at epoch e, the selected checkpoint is:

$$e^* = \arg\max_e \overline{D}_{val}^{(e)}. \tag{5.16}$$

The final reported metrics are computed from this selected checkpoint on the held-out validation cases. Because the same validation set is used for repeated checkpoint selection and descriptive reporting, it must not be interpreted as an independent final test set.

The primary reported segmentation metric is mean Dice across RV, MYO, and LV:

$$\overline{D} = \frac{D_{RV} + D_{MYO} + D_{LV}}{3}. \tag{5.17}$$

At inference, the network returns four logit channels on the resized grid. The class with the largest logit is selected voxel-wise using `argmax`; there is no thresholding, connected-component filtering, hole filling, or other mask post-processing. The reference label is resized with nearest-neighbour interpolation and cast to integer class identifiers. Dice is first computed for each foreground class in each case, then averaged across the 40 validation cases; the three class means are averaged to obtain the reported mean Dice.

The implementation assigns a class Dice of 1 when both the predicted and reference masks for that class are empty in a case; otherwise it applies the unsmoothed foreground Dice expression. This convention is stated because it can affect case-level aggregation. The audited evidence contains 40 per-case rows for each of the six models (720 foreground class scores in total). None of those class scores is exactly 1, so the empty-empty branch did not contribute to the reported aggregate values. The convention is nevertheless retained here because it remains part of the executed metric definition.

For post-hoc uncertainty description, the two ED/ES case values are first averaged within each of the 20 validation patients. A patient-level percentile bootstrap then resamples 20 patients with replacement for 10,000 replicates using audit seed 20260820. This produces 95% intervals for each model's patient-macro mean Dice and for paired Nano-Mamba differences. Resampling at patient level preserves the dependence between the two cases from one patient and between model predictions on the same patient. These intervals were added after recovery of the original per-case CSVs; they are descriptive, not pre-registered hypothesis tests. Because the same 20 patients were used repeatedly for checkpoint selection, the intervals do not convert the validation results into independent test-set estimates.

The study also reports parameter count and inference speed. The reported parameter count sums every tensor returned by `model.parameters()`, including the unused legacy outputs of the gate projection described above. Inference speed uses a random batch-one tensor of shape $1 \times 1 \times 256 \times 256 \times 16$, five warm-up passes, and 30 timed passes with CUDA synchronization. These computational metrics are engineering measurements from one implementation and environment; parameter count is not treated as a direct proxy for speed.

5.10 Hardware and Reproducibility

The experiment used PyTorch (Paszke et al., 2019) and MONAI on a single NVIDIA RTX 4060 GPU with 8GB VRAM. The random seed is fixed to 42 for Python, NumPy, and PyTorch, and CuDNN deterministic settings are enabled. All six result rows are accompanied by 40-case metric tables and 150-epoch training logs; their class averages and selected best epochs reproduce the aggregate result table. The patient split, preprocessing code, model definitions, and evaluation code are retained in the project repository. Deterministic settings improve repeatability within a fixed software and hardware stack, but they do not guarantee bitwise identity across library versions or devices.

5.11 Methodological Summary

The methodology supports a cautious comparison under one patient-level split and one deterministic preprocessing pipeline. It does not provide an independent test set, matched batch sizes, matched-capacity ablations, multi-seed training, augmentation, native-space evaluation, or phase- and pathology-stratified analysis. The model is not evaluated using training-set performance and is not claimed to be a full implementation of the original Mamba selective scan. The experiment therefore addresses whether this particular compact Mamba-inspired bottleneck occupies a competitive accuracy-efficiency region, not whether it is universally superior.

CHAPTER 6: RESULTS

This chapter reports the empirical results obtained from the revised patient-level validation experiment. Unlike the earlier exploratory scripts, the final experiment used a deterministic 80/20 patient-level split of the ACDC training cohort. The split contained 80 training patients and 20 held-out validation patients, corresponding to 160 training cases and 40 validation cases. The best checkpoint for each model was selected using validation mean Dice, and the final quantitative results were reported only on the held-out validation cases.

The objective of this chapter is descriptive rather than argumentative. It reports what the experiment shows before Chapter 7 interprets the implications. This separation is important because the final results are more nuanced than a simple claim of superiority. Nano-Mamba U-Net is not the highest-Dice model in the final experiment. Its value lies in the combination of competitive segmentation performance and the smallest parameter count among the evaluated models.

6.1 Experimental Setting

All models were trained and evaluated using the same deterministic preprocessing and validation protocol. The input volumes were resized to $256 \times 256 \times 16$, intensity-scaled to $[0, 1]$ per volume, and converted to PyTorch tensors using MONAI transforms (Cardoso et al., 2022). Images used trilinear interpolation, labels used nearest-neighbour interpolation, and no data augmentation or mask post-processing was applied. The evaluated anatomical classes were the Right Ventricle (RV), Myocardium (MYO), and Left Ventricle (LV). The primary segmentation metric was the Dice Similarity Coefficient (DSC), computed on the resized grid.

The comparison included a standard 3D U-Net baseline, Attention U-Net, SegResNet16, the proposed Nano-Mamba U-Net, and two ablation variants. SegResNet16 refers to the configuration with `init_filters=16`, which was the version successfully trained under the final patient-level validation protocol. Therefore, the parameter count reported in this chapter is 4.70M for SegResNet16, not the larger 18.80M configuration explored in earlier standalone benchmarking scripts.

For all models, the validation mean Dice was computed as the average of the three foreground classes:

$$\overline{D}_{val} = \frac{D_{RV} + D_{MYO} + D_{LV}}{3}. \tag{6.1}$$

This formulation avoids allowing one strong class, such as LV, to dominate the interpretation. Reporting RV, MYO, and LV separately is necessary because a clinically useful segmentation model should not only perform well on the easiest structure.

6.2 Quantitative Validation Results

Table 6.1 presents the final held-out validation performance and computational statistics. The results show that SegResNet16 achieved the highest validation mean DSC of 86.70%. The proposed Nano-Mamba U-Net achieved a mean DSC of 84.78% with the smallest parameter count among the evaluated models, at 1.46M parameters.

Table 6.1: Held-out patient-level validation results on the ACDC training cohort. Intervals are post-hoc patient-level percentile-bootstrap 95% intervals (10,000 replicates); the same validation set was used for checkpoint selection.

Architecture	RV (%)	MYO (%)	LV (%)	Mean DSC (%)	95% CI (%)	Params (M)	FPS
3D U-Net Baseline	78.35	74.57	89.59	80.83	[77.78, 83.46]	4.81	88.17
Attention U-Net	69.43	72.38	82.53	74.78	[65.37, 82.44]	5.91	22.41
SegResNet16	**84.30**	**82.76**	**93.03**	**86.70**	[84.26, 88.55]	4.70	25.50
Ablation: No Mamba	83.26	81.30	92.37	85.64	[83.08, 87.64]	2.29	28.80
Ablation: Half Mamba	82.41	80.46	91.99	84.95	[81.85, 87.21]	1.64	29.04
Nano-Mamba U-Net	82.11	80.35	91.88	84.78	[82.05, 86.90]	**1.46**	29.09

Compared with the 3D U-Net baseline, Nano-Mamba U-Net improved the mean DSC by 3.95 percentage points while reducing the parameter count from 4.81M to 1.46M. Using the unrounded parameter counts from the result CSV, the reduction relative to 3D

U-Net is:

$$\frac{4.808537 - 1.456325}{4.808537} \times 100\% \approx 69.7\%. \tag{6.2}$$

Compared with SegResNet16, Nano-Mamba U-Net used approximately 69.0% fewer parameters, but its mean DSC was 1.92 percentage points lower. These results indicate that Nano-Mamba U-Net is not the highest-accuracy model in the final experiment; instead, its main advantage lies in parameter efficiency with competitive validation performance.

Table 6.2 reports paired patient-level differences. Positive values favor Nano-Mamba U-Net. Pairing is important because every model was evaluated on the same 20 patients. The intervals are descriptive resampling summaries; no p-values, multiplicity correction, or independent test cohort is claimed.

Table 6.2: Post-hoc paired patient-level Nano-Mamba mean-DSC differences.

Reference model	Nano minus reference (pp)	95% bootstrap CI (pp)
3D U-Net Baseline	+3.95	[+2.83, +5.01]
Attention U-Net	+10.00	[+3.51, +18.29]
SegResNet16	-1.92	[-3.01, -0.88]
Ablation: No Mamba	-0.86	[-1.48, -0.23]
Ablation: Half Mamba	-0.17	[-0.91, +0.55]

6.3 Accuracy-Efficiency Visualization

Since the research question concerns both segmentation accuracy and model efficiency, the results should not be interpreted from Table 6.1 alone. Figure 6.1 visualizes the relationship between validation mean Dice and trainable parameter count. A model positioned higher on the vertical axis has stronger Dice performance, while a model positioned further left has fewer parameters.

This visualization makes the central interpretation clearer. SegResNet16 is the accuracy leader and should be acknowledged as such. Nano-Mamba U-Net is not located at the top of the graph, but it appears on the left side because it has the smallest parameter count.

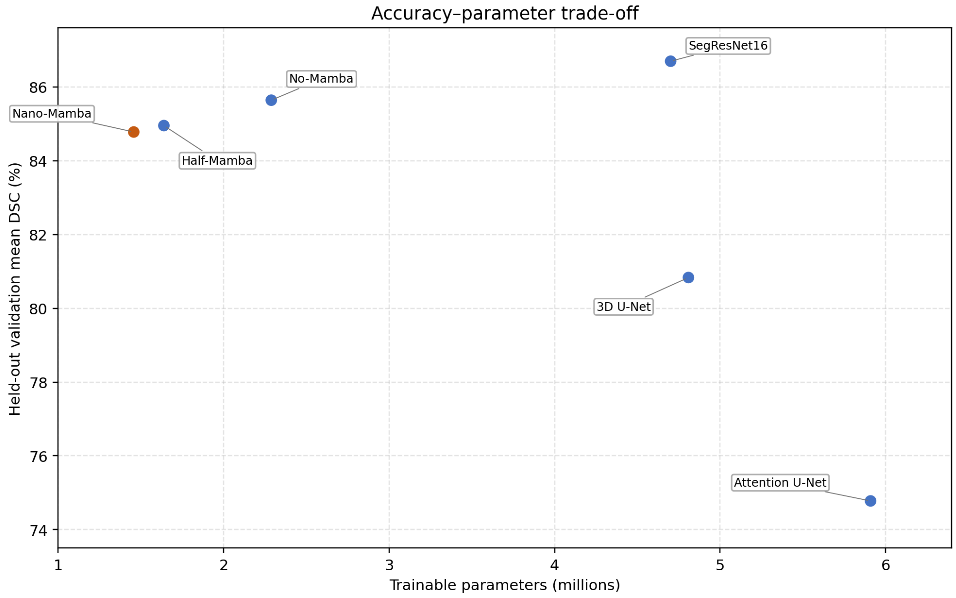

Figure 6.1: Accuracy-parameter trade-off generated directly from the audited summary CSV. SegResNet16 obtains the highest held-out validation Dice score, while Nano-Mamba U-Net occupies the lowest-parameter region.

The result therefore supports an efficiency-oriented contribution rather than an absolute accuracy claim.

Figure 6.2 visualizes the relationship between validation mean Dice and inference speed. The speed result should be interpreted carefully because FPS depends on implementation details, hardware, batch size, input resolution, and measurement protocol. Therefore, FPS is reported as a practical engineering indicator rather than as a universal property of the architecture.

6.4 Model-by-Model Interpretation

The 3D U-Net baseline achieved 80.83% mean DSC. This result is important because it provides a conventional volumetric segmentation reference point. Nano-Mamba U-Net improved over this baseline on all three structures, suggesting that the proposed compact architecture is not merely smaller but also more accurate than this particular 3D U-Net configuration under the final split.

Attention U-Net achieved the lowest mean DSC among the evaluated models, at 74.78%.

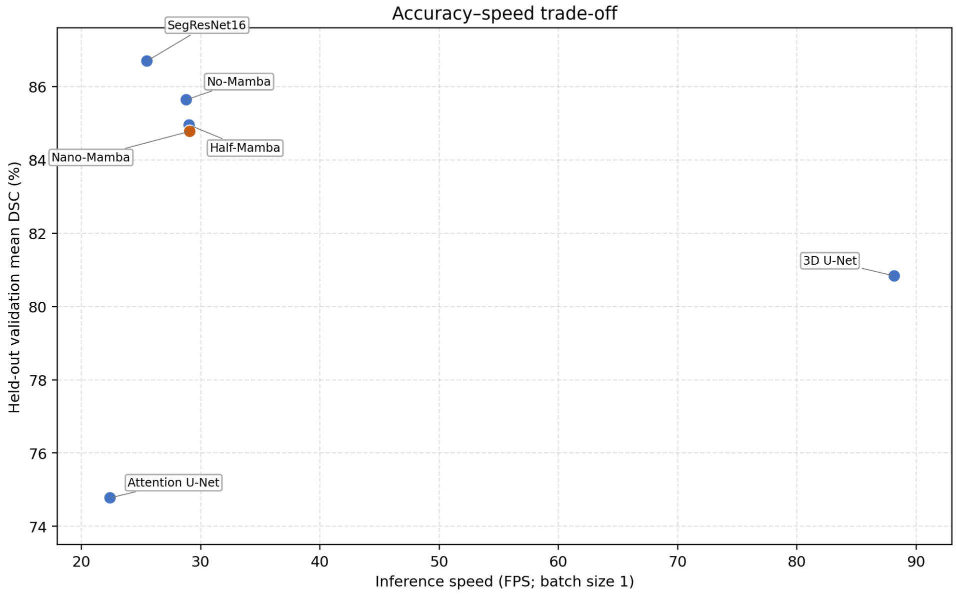

Figure 6.2: Accuracy-speed trade-off generated directly from the audited summary CSV. The figure provides an engineering view of held-out validation Dice relative to batch-one inference speed in the recorded experiment.

This does not mean attention gates are generally ineffective. It means that, under the implemented settings and validation split, this Attention U-Net configuration did not train or generalize as strongly as the other models. This result also demonstrates why it is unsafe to assume that a more complex architectural concept necessarily improves performance.

SegResNet16 achieved the highest mean DSC, with the best RV, MYO, and LV scores. This suggests that residual convolutional modeling remains very strong for this dataset. Since SegResNet16 outperformed Nano-Mamba U-Net by 1.92 percentage points, the thesis must not describe Nano-Mamba U-Net as the best model by Dice. Instead, SegResNet16 should be acknowledged as the accuracy leader in the final experiment.

The No-Mamba ablation achieved 85.64% mean DSC, which is 0.86 percentage points higher than Nano-Mamba U-Net. The paired post-hoc interval for Nano minus No-Mamba is $[-1.48, -0.23]$ percentage points. This is an important negative result because it prevents over-claiming the Mamba-inspired module. The No-Mamba design is not parameter matched and uses 2.29M parameters compared with Nano-Mamba's 1.46M, so the comparison does not isolate sequence gating as the cause of the difference. It shows only that the executed No-Mamba architecture obtained higher Dice on this selected split while Nano-Mamba retained the smaller reported parameter count.

The Half-Mamba variant achieved 84.95% mean DSC, 0.17 percentage points above Nano-Mamba U-Net. The paired interval for Nano minus Half-Mamba is $[-0.91, +0.55]$ percentage points and crosses zero, so the available case-level evidence does not resolve this small ordering reliably. A matched-capacity, multi-seed experiment would be required to study bottleneck size more reliably.

6.5 Class-Specific Performance

Across most models, LV segmentation achieved the highest Dice score, while MYO segmentation was generally more difficult. This pattern is consistent with the anatomical properties of cardiac MRI: the LV cavity is usually more clearly defined, whereas the myocardium is a thinner ring-like structure affected by papillary muscles, partial volume effects, and boundary ambiguity.

For Nano-Mamba U-Net, the LV Dice was 91.88%, the RV Dice was 82.11%, and the MYO Dice was 80.35%. The gap between LV and MYO is:

$$91.88\% - 80.35\% = 11.53\%. \tag{6.3}$$

This difference is clinically plausible because the myocardium is more difficult to delineate consistently. A small absolute boundary shift can affect a thin myocardial ring more strongly than a larger ventricular cavity. Therefore, the lower MYO score should be interpreted as a known structural challenge rather than simply as a random model failure.

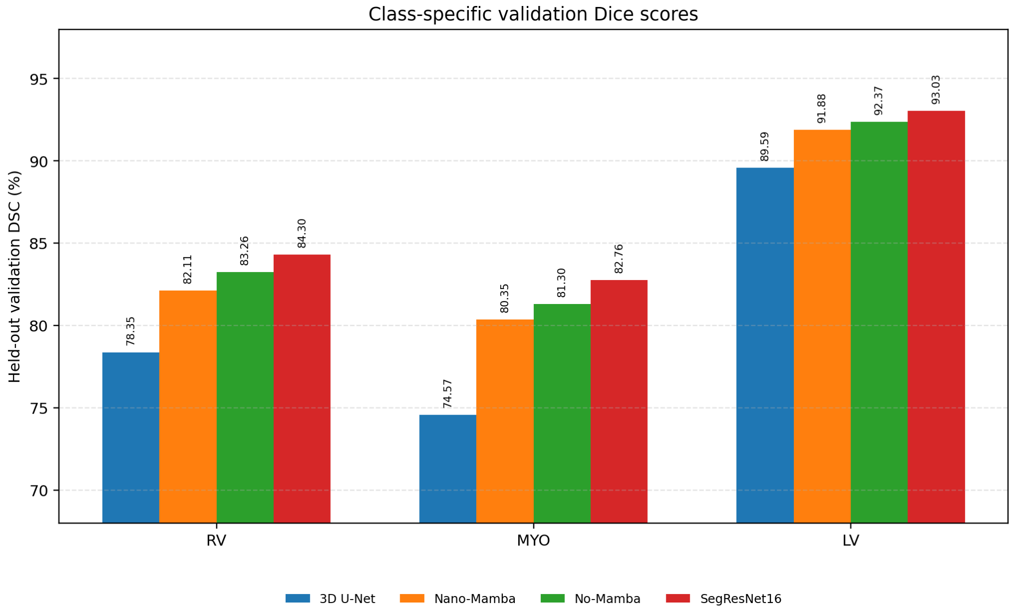

Figure 6.3: Class-specific held-out validation Dice scores generated directly from the audited summary CSV. Nano-Mamba U-Net improves over the evaluated 3D U-Net baseline, while SegResNet16 achieves the strongest scores.

6.6 Checkpoint Selection Evidence

The experiment logs contain 150 contiguous epoch rows for every model. For each log, the maximum recorded validation mean DSC occurs at exactly the best epoch in the aggregate table: epochs 105, 121, 134, 149, 62, and 112 for 3D U-Net, Nano-Mamba, No-Mamba, Half-Mamba, Attention U-Net, and SegResNet16, respectively. Figure 6.4 therefore shows the recorded raw validation traces rather than hand-entered points.

The learning curves also show why validation-based checkpoint selection was necessary. By epoch 150, the validation score was 0.29 percentage points below the selected maximum for both Nano-Mamba and 3D U-Net, 0.43 points lower for SegResNet16, 0.56–0.62 points lower for the two ablations, and 3.05 points lower for Attention U-Net. Training loss nevertheless continued to decrease for all six models. The larger Attention U-Net gap is the clearest sign of late-epoch overfitting or instability in the recorded runs; it does not imply that attention mechanisms are generally inferior.

The per-case tables identify a shared difficult case. `patient049_frame11` is the

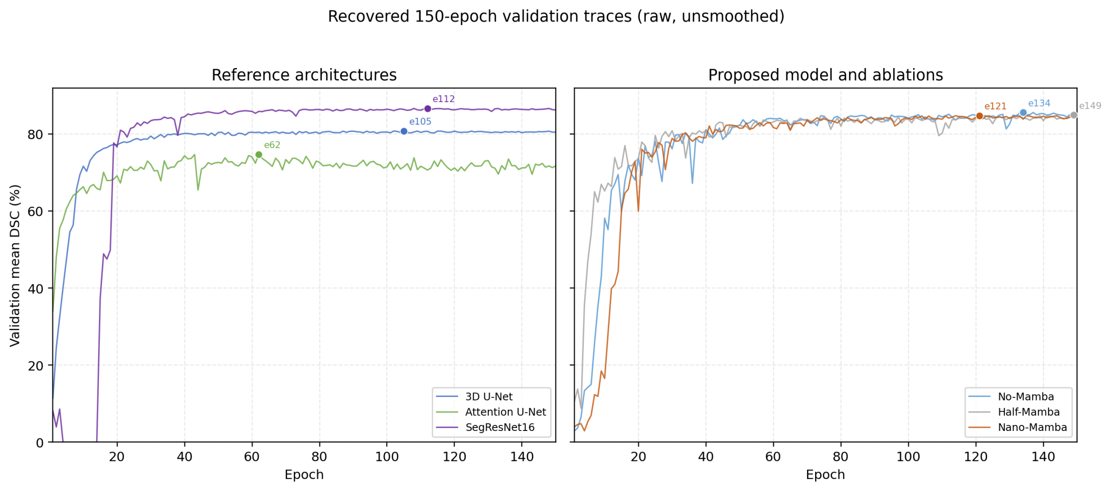

Figure 6.4: Raw, unsmoothed 150-epoch validation mean-DSC traces. Markers identify the best validation checkpoint selected for each model. The figure is generated directly from the six experiment-log CSVs.

lowest-mean-Dice case for Nano-Mamba, 3D U-Net, SegResNet16, No-Mamba, and Half-Mamba. Nano-Mamba obtains 57.86% mean DSC on this case, driven mainly by low RV Dice of 26.04%. Attention U-Net's worst case is `patient042_frame16`, at 16.38% mean DSC. Because no pathology label or visual overlay is used in this analysis, these numerical failures should not be assigned an anatomical cause without inspecting the corresponding validation images.

No qualitative prediction panel is included in the final thesis. Any future panel should be generated from a validation patient and the selected Nano-Mamba checkpoint, and should clearly state that it illustrates one case rather than proving general performance.

CHAPTER 7: DISCUSSION AND LIMITATION

This chapter interprets the final validation results and explains the implications for the proposed Nano-Mamba U-Net. The discussion is intentionally conservative: the model is evaluated as a lightweight segmentation architecture, not as an absolute state-of-the-art solution. This distinction is necessary because the final results show a trade-off rather than a universal win.

7.1 Accuracy-Efficiency Trade-off

The strongest model by Dice was SegResNet16, with a validation mean DSC of 86.70%. Nano-Mamba U-Net achieved 84.78%, which is 1.92 percentage points lower. Therefore, the final experiment does not support the claim that Nano-Mamba U-Net is the most accurate model overall.

However, Nano-Mamba U-Net achieved this performance using only 1.46M parameters, the smallest parameter count among all evaluated models. Relative to the 3D U-Net baseline, Nano-Mamba improved mean DSC by 3.95 percentage points while using substantially fewer parameters. This supports the central contribution of the thesis: Nano-Mamba U-Net provides a compact architecture with competitive validation performance, suitable for exploring lightweight cardiac segmentation.

The trade-off can be interpreted through the relationship between model size and segmentation accuracy. If D denotes mean Dice and P denotes parameter count, then an informal efficiency ratio can be written as:

$$E = \frac{D}{P}. \tag{7.1}$$

This ratio should not be treated as a clinical metric, but it helps illustrate the engineering perspective. Nano-Mamba U-Net has a lower Dice than SegResNet16 but also a much lower parameter count. Therefore, its value is not that it dominates every metric, but that it occupies a favorable lightweight region of the accuracy-efficiency space.

7.2 Interpretation of Ablation Results

The ablation results require careful interpretation. The No-Mamba variant achieved 85.64% mean DSC, slightly higher than Nano-Mamba U-Net. This indicates that the current lightweight Mamba-inspired bottleneck does not consistently improve Dice over a stronger convolutional bottleneck under the final validation split. Therefore, the Mamba-inspired module should not be presented as conclusively superior in segmentation accuracy.

At the same time, Nano-Mamba U-Net reduced the parameter count compared with the No-Mamba variant, from 2.29M to 1.46M, while maintaining a close mean DSC. This suggests that the bottleneck is more valuable as a parameter-efficiency design than as a guaranteed accuracy-improving component. The Half-Mamba variant achieved 84.95%, which is close to Nano-Mamba, further showing that channel allocation and bottleneck capacity affect the accuracy-efficiency balance.

A possible explanation is that the compressed bottleneck operates on a very low-resolution representation. At this stage, the network may already have lost some fine anatomical boundary information. A sequence module can mix information across the compressed representation, but it cannot recover details that were not preserved by the encoder. This explains why the Mamba-inspired module may help compactness without guaranteeing the best Dice. Future work should investigate whether placing sequence modules at multiple resolutions would improve this balance.

7.3 Class-Specific Observations

LV segmentation was the easiest class across the main models, while MYO segmentation was more difficult. This pattern is clinically reasonable because MYO is a thinner structure and its boundary can be affected by papillary muscles and partial volume effects. RV segmentation also remains challenging because the right ventricle has a more variable shape than the left ventricle, especially in pathological cases.

The class-specific findings support the need for reporting RV, MYO, and LV separately. If only mean Dice were reported, the relatively strong LV score could hide weaker MYO performance. Since cardiac measurements such as myocardial mass depend on MYO segmentation, MYO accuracy cannot be treated as a secondary detail. A future version of the project should therefore include additional metrics such as Hausdorff Distance or Average Surface Distance to better evaluate boundary quality.

7.4 Relevance to Cardiac Motion Analysis

The thesis title includes cardiac motion analysis, but the implemented experiment is segmentation-based. The model does not estimate a dense displacement field, optical flow, temporal deformation map, or temporal consistency. Instead, it segments labelled ED and ES 3D volumes as separate cases. In the input tensor, D denotes anatomical slice depth rather than time. The resulting masks could support later computation of phase-specific volumes, but this experiment does not calculate ejection fraction or other motion-related clinical indices.

This limitation does not invalidate the project, but it narrows the claim. The correct interpretation is that Nano-Mamba U-Net contributes to segmentation as a prerequisite for cardiac functional analysis. A stronger future version would process complete 4D cine-MRI sequences and evaluate temporal consistency directly.

7.5 Limitations

Several limitations must be acknowledged.

1. **Internal validation and checkpoint reuse:** The final experiment used a deterministic patient-level validation split of the ACDC training cohort. The same validation patients were inspected across epochs to select each checkpoint and were then used for the descriptive result table. This is not an independent test set and does not represent official ACDC challenge performance.
2. **Single split, seed, and post-hoc uncertainty:** Only seed 42 and one 80/20 patient split were reported. The available per-case CSVs support patient-level bootstrap intervals and paired differences, but these were computed after the experiment and use the same validation patients involved in checkpoint selection. They do not replace multi-seed training or an independent test cohort.
3. **Mamba-inspired implementation:** The implemented bottleneck is inspired by Mamba-style sequence modeling but is not a full reproduction of the original hardware-aware selective scan implementation. This distinction is important because the thesis must not attribute the full original Mamba mechanism to the implemented model.
4. **Legacy projection parameters:** The historical gate projection produces $2d_{state}+1$ channels but uses only the first. The reported parameter count includes the unused outputs. They were retained to preserve checkpoint compatibility, but a redesigned block would need retraining.
5. **No explicit temporal modeling:** The implemented task segments ED and ES 3D MRI volumes independently. It performs neither dense motion tracking nor sequence-level consistency analysis, and the depth dimension is not time.
6. **Non-matched training and ablations:** Memory constraints required batch size 1 for

Attention U-Net and SegResNet16 and batch size 2 for the other models. Optimizer updates per epoch and normalization choices therefore differ across model families. The No-Mamba ablation also has more parameters and a structurally different bottleneck. These factors limit causal attribution of performance differences to one architectural mechanism.

7. **Array-size preprocessing and no augmentation:** The pipeline resizes arrays to $256 \times 256 \times 16$ but does not explicitly canonicalize orientation, preserve a common physical voxel spacing, crop around the heart, or inverse-resample predictions to native geometry. No training-time data augmentation is used. Dice is therefore measured on a deterministic resized grid, and geometric robustness is not tested.
8. **Metric convention:** An empty prediction and empty reference for one class are assigned Dice 1 in the executed metric. None of the 720 audited foreground class scores equals 1, establishing that this branch did not affect the reported case table; the convention could still matter in future datasets.
9. **No phase- or pathology-stratified analysis:** Although ACDC contains ED/ES phase information and pathology labels, the final experiment pools the two cases per validation patient and reports aggregate performance. It does not establish whether performance differs systematically between ED and ES or among diagnostic groups.
10. **Limited external validation:** The model was not evaluated on a separate multi-center external dataset. Generalization to other scanners and institutions remains future work.
11. **Limited metrics and qualitative evidence:** The main reported metric is Dice; no Hausdorff Distance, Average Surface Distance, calibration metric, or blinded qualitative review is available. FPS remains a hardware- and implementation-specific engineering indicator rather than a universal benchmark.

These limitations define the thesis boundary. The case rows and training logs are internally consistent with the numerical table and post-hoc intervals, but they do not remove the design limitations above. Clinical deployment, temporal motion analysis, and external generalization remain outside scope.

7.6 Summary of Findings

The final validation results reposition Nano-Mamba U-Net as a lightweight, parameter-efficient segmentation model. It outperformed the 3D U-Net baseline in mean DSC and used fewer parameters than all evaluated alternatives. Nevertheless, SegResNet16 achieved the highest absolute Dice score, and the No-Mamba ablation slightly exceeded Nano-Mamba in mean DSC. These findings support a balanced conclusion: the proposed model is competitive and efficient, but not definitively superior in accuracy.

CHAPTER 8: CONCLUSION AND FUTURE WORK

This chapter concludes the thesis by summarizing the final experimental findings, answering the research questions, and identifying future directions. The conclusion is based on the revised patient-level validation experiment reported in Chapter 6 and the interpretation provided in Chapter 7. The purpose of this chapter is not to overstate the performance of the proposed model, but to clearly identify what was achieved, what was not achieved, and what remains open for future research.

8.1 Restatement of the Research Aim

The aim of this thesis was to design and evaluate a lightweight Mamba-inspired U-Net architecture for cardiac MRI segmentation. The motivation came from a practical research problem: cardiac segmentation models should not only achieve high Dice scores, but should also be computationally manageable and evaluated under a reproducible validation protocol. The final implementation, Nano-Mamba U-Net, integrates a sequence-processing bottleneck into a 3D U-Net-style encoder-decoder network. Compressed volumetric features are serialized into a one-dimensional sequence, processed by a lightweight gated sequence module inspired by Mamba, and reshaped back into the 3D feature space.

This thesis does not claim that Nano-Mamba U-Net is a complete implementation of the original Mamba selective scan algorithm. It also does not claim official ACDC test-set performance or clinical deployment readiness. Its contribution is more specific and more defensible: it implements and evaluates a compact Mamba-inspired bottleneck for cardiac MRI segmentation, compares it against several baselines and ablations, and reports the resulting accuracy-efficiency trade-off under a patient-level validation split.

8.2 Summary of Research Contributions

The main contributions are summarized as follows:

1. **Architecture implementation:** A Nano-Mamba U-Net architecture was implemented for 3D cardiac MRI segmentation of RV, MYO, and LV. The model uses a compact encoder-decoder structure and a Mamba-inspired bottleneck at the compressed latent representation. This contribution is architectural integration rather than invention of the original U-Net or Mamba methods.
2. **Rigorous validation protocol:** A deterministic patient-level 80/20 split was used to avoid leakage between training and validation patients. The final experiment used 160 training cases and 40 held-out validation cases from the ACDC training cohort. This corrected the weakness of earlier exploratory scripts that were not sufficiently strict about held-out validation.
3. **Accuracy-efficiency analysis:** Nano-Mamba U-Net achieved 84.78% mean DSC with a post-hoc patient-bootstrap 95% interval of 82.05–86.90% and 1.46M parameters. It improved over the 3D U-Net baseline in Dice while using fewer parameters. However, it did not achieve the highest Dice score overall; SegResNet16 obtained the highest validation mean DSC of 86.70%.
4. **Transparent ablation findings:** The No-Mamba ablation achieved 85.64% mean DSC, slightly higher than Nano-Mamba U-Net. This shows that the current Mamba-inspired bottleneck should be interpreted mainly as a parameter-efficiency design rather than a conclusively superior accuracy mechanism.
5. **Thesis-level scientific correction:** The final thesis deliberately avoids unsupported claims such as absolute state-of-the-art superiority, complete Mamba implementation, or official clinical readiness. This is important because a rigorous research project should report evidence accurately even when the results are mixed.

8.3 Answers to Research Questions

Research Question 1: How can compressed 3D cardiac MRI feature maps be serialized and processed by a Mamba-inspired bottleneck within a U-Net-style segmentation architecture?

The project answers this question through the Nano-Mamba U-Net design. A 3D encoder first compresses the MRI volume into a lower-resolution latent feature map. This tensor is reshaped from $B \times C \times H \times W \times D$ into a one-dimensional sequence of HWD tokens, processed by a lightweight Mamba-inspired gated sequence block, and then reshaped back into volumetric form before the decoder reconstructs the segmentation mask. This demonstrates a practical way to insert sequence modeling into a 3D U-Net-style segmentation pipeline.

The significance of this design is that sequence processing is applied at the bottleneck, where the spatial resolution is already reduced. This keeps the computational cost manageable. If the sequence module were applied directly to the full-resolution input volume, the token length would be much larger and the memory burden would increase substantially. Therefore, the bottleneck placement reflects an engineering compromise between contextual mixing and computational feasibility.

However, this design also has a limitation. Because the bottleneck operates after several down-sampling steps, some fine boundary information may already have been compressed by the encoder. The sequence module can mix the remaining latent features, but it cannot fully recover details that were lost earlier. This helps explain why the architecture is compact and competitive but not the highest-performing model in the final validation experiment.

Research Question 2: Does the proposed Nano-Mamba U-Net achieve competitive held-out validation Dice compared with 3D U-Net, Attention U-Net, SegResNet, and

ablation variants?

The answer is partially yes. Nano-Mamba U-Net achieved 84.78% mean DSC, outperforming the 3D U-Net baseline (80.83%) and Attention U-Net (74.78%). The paired post-hoc intervals for Nano minus these two models lie above zero. Nano-Mamba was lower than SegResNet16 (86.70%) and No-Mamba U-Net (85.64%), with paired intervals below zero. Half-Mamba U-Net obtained 84.95%, but its paired interval against Nano crosses zero. Therefore, Nano-Mamba U-Net achieved competitive validation performance, but it should not be described as the top-performing model by Dice or as statistically validated on an independent cohort.

This result is scientifically useful because it avoids a simplistic conclusion. If only the comparison against 3D U-Net were considered, Nano-Mamba U-Net would appear clearly better. If only the comparison against SegResNet16 were considered, Nano-Mamba U-Net would appear weaker. The complete interpretation requires considering all models together. The proposed model improves over some baselines, remains close to some ablations, and trails the strongest residual segmentation model.

The final answer to this research question is therefore conditional: Nano-Mamba U-Net is competitive in held-out validation performance, but it is not the strongest model by absolute Dice. This conclusion is more cautious than the original thesis draft, but it is more defensible for academic assessment.

Research Question 3: What accuracy-efficiency trade-off is observed when comparing Nano-Mamba U-Net with heavier or non-Mamba alternatives?

Nano-Mamba U-Net achieved the smallest reported parameter count among all evaluated models, with 1.46M parameters. Using the unrounded CSV values, it used approximately 69.7% fewer reported parameters than the 3D U-Net baseline and approximately 69.0% fewer than SegResNet16. The reported count includes unused legacy gate-projection outputs, so it describes serialized trainable tensors rather than effective parameters only. The trade-off is that this compactness came with lower Dice than SegResNet16 and the No-Mamba ablation. The final interpretation is that Nano-Mamba U-Net offers a lightweight design, but not the highest absolute segmentation accuracy.

This trade-off is central to the thesis. The project should not be judged only by whether it wins a single Dice ranking. A lightweight model can still be meaningful when it provides acceptable accuracy with fewer trainable parameters. At the same time, parameter efficiency alone is not enough; the segmentation quality must remain clinically and technically reasonable. The final result suggests that Nano-Mamba U-Net sits in a useful middle ground: it is smaller than all evaluated models and more accurate than the basic 3D U-Net baseline, but it does not replace SegResNet16 as the accuracy leader.

8.4 Implications of the Findings

The findings have three main implications. First, U-Net-style architectures remain strong foundations for cardiac MRI segmentation. The success of SegResNet16 and the No-Mamba ablation shows that convolutional and residual components are still highly effective on this dataset. This prevents the thesis from making an exaggerated claim that new sequence-inspired modules automatically outperform established convolutional methods.

Second, lightweight sequence-inspired modules remain worth exploring. Nano-Mamba U-Net achieved competitive validation performance with the lowest parameter count. This suggests that compact sequence processing at the bottleneck can be integrated into medical segmentation architectures without causing severe performance collapse. The result is not a final proof of superiority, but it is a reasonable experimental basis for further study.

Third, the validation protocol matters as much as the architecture. The revised patient-level split produced more conservative and more credible results than earlier exploratory evaluations. This is an important lesson for medical AI research: high Dice scores are less meaningful if the evaluation protocol is not strict. A lower but properly validated score is more useful than an inflated score obtained from a weak experimental design.

8.5 Future Work

Several directions can strengthen this work further.

8.5.1 Evaluation on External and Official Test Sets

Future experiments should evaluate the model on the official ACDC test set or on independent external cardiac MRI datasets. This would provide stronger evidence of generalization beyond the internal patient-level validation split. External validation is particularly important in medical imaging because scanner vendor, acquisition protocol, image resolution, and patient population can all affect model performance.

8.5.2 Pathology-Specific Validation

The ACDC dataset includes diagnostic categories such as NOR, DCM, HCM, MINF, and ARV. A future version should parse the diagnostic metadata and report pathology-specific Dice scores. This would allow a more clinically meaningful analysis of where the model performs well and where it fails. For example, RV segmentation may be more difficult in abnormal right ventricle cases, while myocardial segmentation may be affected by hypertrophic cardiomyopathy or myocardial infarction. Such subgroup analysis should be generated from actual patient metadata rather than assumed or manually invented.

8.5.3 Additional Boundary-Based Metrics

The current thesis focuses mainly on Dice Similarity Coefficient. Dice is suitable for overlap-based evaluation, but it does not fully describe boundary quality. Future work should report Hausdorff Distance, Average Surface Distance, or related boundary metrics.

This is especially relevant for myocardium segmentation, where small boundary errors can affect clinical measurements even when overlap scores appear acceptable.

8.5.4 Closer Integration with Full Mamba/SSM Implementations

The current implementation is Mamba-inspired rather than a full selective scan implementation. Future work could compare the lightweight PyTorch block against an official or optimized Mamba implementation to determine whether a more faithful state space module improves cardiac segmentation. Such a comparison would clarify whether the current result is limited by the simplified bottleneck or by the broader suitability of sequence modeling for compressed cardiac features.

8.5.5 Multi-Scale Sequence Modeling

The present model applies the Mamba-inspired module only at the bottleneck. Future work could place sequence modules at multiple encoder or decoder resolutions. This may help preserve finer boundary information while still enabling wider contextual mixing. However, multi-scale sequence modeling would also increase memory usage and implementation complexity, so it should be evaluated carefully rather than assumed to be superior.

8.5.6 Scaling Toward True 4D Cardiac Motion Modeling

The current study processes labelled ED and ES 3D cardiac volumes independently; the tensor depth axis represents slices rather than time. A natural extension is to process full 4D cine-MRI sequences, where the model receives multiple temporal phases from the cardiac cycle. This would make the title's motion-analysis direction explicit by modeling temporal consistency across the entire cardiac cycle rather than only providing phase-specific segmentation masks.

True 4D modeling would require a different evaluation design. The model would need to preserve temporal consistency, avoid unrealistic frame-to-frame segmentation jumps, and support quantitative motion-related indices. This remains outside the implemented scope of the current thesis, but it is the most direct future direction for connecting the present segmentation work to cardiac motion analysis.

8.6 Concluding Remarks

This thesis demonstrates that a compact Mamba-inspired bottleneck can be integrated into a 3D U-Net-style architecture for cardiac MRI segmentation. The final experiment does not show absolute state-of-the-art accuracy, but it does show that Nano-Mamba U-Net provides competitive validation performance with the smallest parameter count among the evaluated models. The main value of the project is therefore its exploration of lightweight sequence-inspired design for cardiac segmentation under practical hardware constraints.

The most important conclusion is that the proposed model should be understood as an accuracy-efficiency trade-off model. It is not a universal replacement for stronger residual segmentation networks, and it is not a complete clinical motion-analysis system. It contributes an implementation and an evidence-linked evaluation of a compact Mamba-inspired segmentation architecture. The present evidence supports the internal six-model comparison, while independent evaluation, multi-seed training, matched-capacity ablations, phase- and pathology-specific analysis, native-space preprocessing, boundary-based metrics, and true 4D temporal modeling would extend this work into a stronger framework for cardiac image analysis.

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